

Electromechanical System: Construction and Working

1. Introduction

An electromechanical system combines electrical and mechanical components to convert electrical energy into mechanical motion or vice versa. These systems are key elements in robotics, automation, industrial machines, and household appliances.

2. Major Components of an Electromechanical System

A. Electrical Components:

- Power supply (AC/DC)
- Controllers (microcontrollers, PLCs, driver circuits)
- Actuators (motors, solenoids, relays)

B. Mechanical Components:

- Gears, shafts, pulleys
- Bearings, linkages
- Mechanical load and housing

C. Optional Feedback Components:

- Encoders
- Limit switches
- Force and temperature sensors

3. Working Principle

1. Electrical power is delivered from a power supply.
2. Controllers send signals to actuators based on programmed logic.
3. Actuators convert electrical energy into magnetic fields.
4. Magnetic fields produce mechanical force or rotation.
5. Mechanical elements transmit motion to the desired load.
6. Sensors provide feedback for closed-loop control when precision is needed.

4. Example: Motor-Driven Electromechanical System

- A controller applies voltage to a motor.
- The motor generates torque through electromagnetic induction.
- Gears adjust speed and torque levels.
- The load receives controlled mechanical motion.
- Feedback sensors maintain accuracy.

5. Applications

- Robotics and automation
- CNC machines
- Automotive systems (ABS, power windows)
- Household appliances
- Industrial actuators and conveyors

6. Conclusion

Electromechanical systems are essential for modern engineering. Their combined electrical and mechanical construction enables precise and efficient motion, making them suitable for a wide range of applications.